

Scoreboard Data Extraction from Video

Computer Vision Engineer - Round 1 Assessment

Candidate Name	_____
Role	Computer Vision Engineer
Input	bowling_scoreboard.mp4
Technologies	Python, OpenCV, NumPy, Tesseract OCR
Output	Timestamped scoreboard data + annotated demo video

1. Objective

Develop a Computer Vision pipeline that processes the supplied bowling scoreboard video, detects the scoreboard when visible, identifies the active player, and extracts the active player's cumulative TTL score from video frames.

2. Methodology

Step	Method
1	OpenCV reads the video frame by frame.
2	Template matching checks whether the scoreboard is visible, so animation frames are skipped.
3	HSV color segmentation detects the yellow-highlighted active row.
4	The player name is recognized using a yellow-text template, with Tesseract OCR as fallback.
5	The TTL region is isolated and the white digit shapes are verified against reference templates from the supplied video.
6	Only changes in player/TTL are stored to a timestamped CSV.

3. Input Video / Frame

The following frame is taken directly from the supplied bowling scoreboard video. The scoreboard contains four player rows and a TTL column on the right.

6 TARUN											
	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -							0
	15	20	27	31							31
V	8 -	3 -	7 1	8 1							0
	8	11	19	28							28
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -								0
	7	25	33								33
2.5											

The active player row is highlighted in yellow. This visual property is used for active-row detection.

4. Detected Scoreboard and Extracted Data

The annotated output highlights the active row and displays the extracted player name and TTL.

Extracted: **ABDISH** | TTL: 31

	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -							0
	15	20	27	31							31
V	8 -	3 -	7 1	8 1							0
	8	11	19	28							28
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -	3 4							0
	7	25	33	40							40

Scoreboard detected | OpenCV + DCR/template verification

Extracted: **VISHAL** | TTL: 28

	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -	X						0
	15	20	27	31							41
V	8 -	3 -	7 1	8 1							0
	8	11	19	28							28
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -	3 4							0
	7	25	33	40							40

Scoreboard detected | OpenCV + DCR/template verification

5. Code Execution

The project was executed on the supplied video using the following command. The run generated a CSV file and an annotated demo video.

```
$ python main.py --video bowling_scoreboard.mp4 --output output/scoreboard_results.csv --demo output/demo_output.mp4
```

Detected updates: 5

CSV output: scoreboard_results.csv

Demo video: output/demo_output.mp4

Sample extracted records:

0.0 | T | TARUN | 33

7.5 | T | TARUN | 36

28.5 | J | JAGDISH | 31

36.0 | V | VISHAL | 28

52.5 | V | VISHAL | 37

The implementation is contained in main.py. The project also includes requirements.txt, demo.py and README.md with setup and run instructions.

6. Extracted Scoreboard Results

Representative changes detected from the supplied video:

Timestamp (s)	Active Row	Player	TTL
0.0	T	TARUN	33
7.5	T	TARUN	36
26.5	J	JAGDISH	31
36.0	V	VISHAL	28
52.5	V	VISHAL	37

7. Submission Links

GitHub Repository: _____

Demo Video: _____

Conclusion: The proposed pipeline successfully detects scoreboard frames, identifies the active player, and extracts representative TTL updates from the supplied video while ignoring non-scoreboard animation frames.